MA388 Sabermetrics: Lesson 28

Home Run Hitting I

```r
library(tidyverse)
library(knitr)
library(baseballr)
```

Lesson Objectives

1. Fit a GAM to model home run probability as a smooth function of launch speed and launch angle.
2. Use the 2015 model to predict home run probability on 2017 batted balls.
3. Simulate 1,000 seasons to compare predicted vs. actual home run counts in 2017.
4. Draw conclusions about whether changes in the baseball or changes in hitters better explain increased HR rates.

Review

In a logistic model for hit probability, is it appropriate to use a linear function of launch angle and exit velocity? Explain.

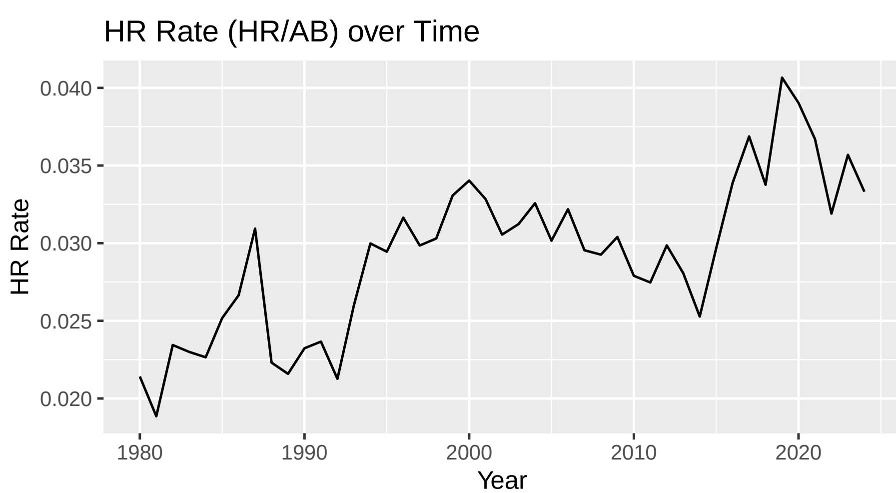
In the DiMaggio streak example, explain how and why we used simulation.

What's with all these home runs?

Video

Here is what they are talking about:

```
library(tidyverse)
library(Lahman)

HRrates <- Teams |>
  filter(yearID > 1979) |>
  group_by(yearID) |>
  summarize(HR = sum(HR),
            AB = sum(AB)) |>
  mutate(rate = HR/AB)

HRrates |>
  ggplot(aes(x = yearID, y = rate)) +
  geom_line() +
  labs(title = "HR Rate (HR/AB) over Time", x = "Year", y = "HR Rate")
```



Some potential explanations:

1. **Explanation 1.** The additional home runs are due to changes in the baseball.
2. **Explanation 2.** The additional home runs are due to changes in the hitters.

Today, we are going to assess which explanation is more compatible with the data.

Here is the approach:

1. Fit a model for home run probability as a function of launch speed and angle **in the 2015 season**.
2. Use the model to estimate the predicted probability of a home run on each ball put in play **in the 2017 season**.
3. Use the estimated probabilities to simulate 1,000 seasons.
4. Compare the home runs in the simulated seasons to the observed home runs in 2017.

What does this tell us?

If the observed rate in 2017 is consistent with the simulated results, which explanation does this support? Why?

What other factors should we consider?

Fit a model for home run probability in 2015.

```
library(tidyverse)
library(baseballr)
library(lubridate)
```

Obtain all pitches in 2015 in which the ball was put in play.

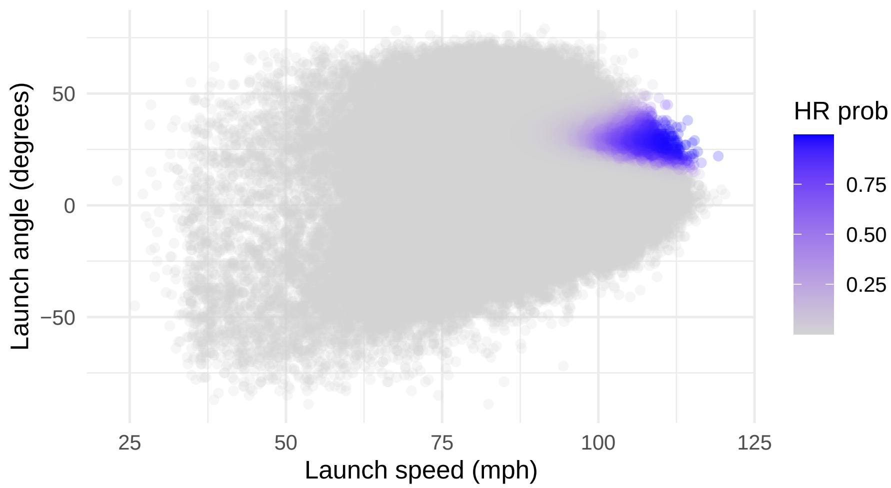
```r
hit_into_play2015 <- read_csv("statcast_2015_bip.csv") |>
  filter(launch_speed < 150,
         launch_speed > 0,
         launch_angle < 360)
```

Fit a model for home run probability using the 2015 data.

```r
library(mgcv)
fit <- hit_into_play2015 |>
  # sample_frac(size=.5) |>
  gam(events == "home_run" ~ s(launch_speed, launch_angle),
      family = binomial,
      data = _)
```

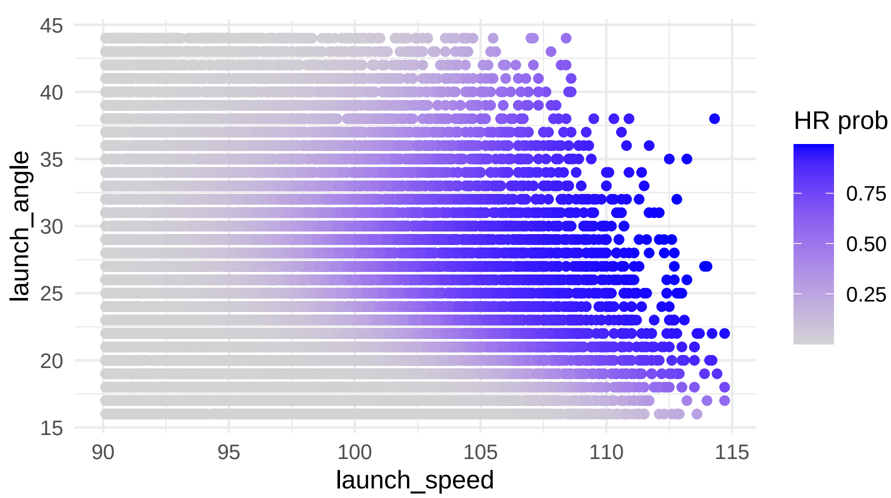
Visualize the predictions.

```r
library(broom)
hit_into_play2015 <- fit |>
  augment(type.predict = "response",
          newdata = hit_into_play2015) |>
  rename(HRprob = .fitted)

hr_prob_2015_fig <-
  hit_into_play2015 |>
  ggplot(aes(x = launch_speed, y = launch_angle, color = HRprob)) +
  geom_point(alpha = 0.2) +
  scale_color_gradient("HR prob", low = "lightgray", high = "blue") +
  theme_minimal()+
  labs(x='Launch speed (mph)',y='Launch angle (degrees)')

hr_prob_2015_fig
```



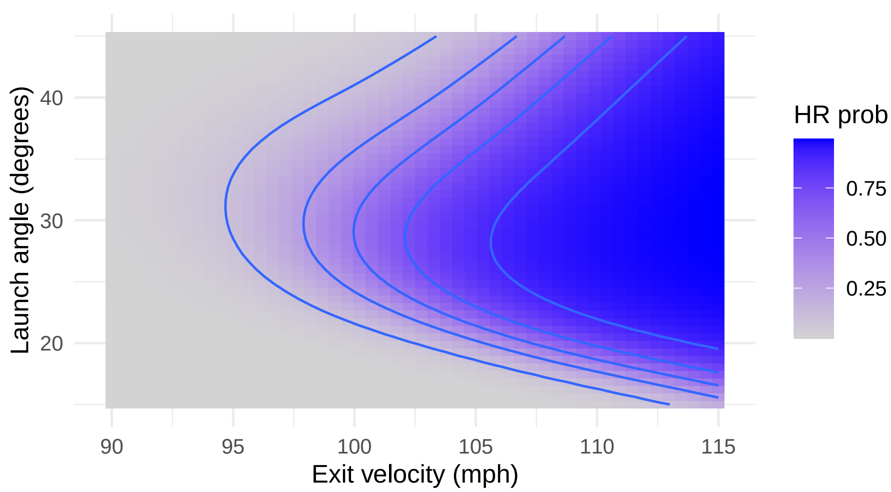
Let's zoom in.

```
hit_into_play2015 |>
  filter(launch_speed > 90, launch_speed < 115,
         launch_angle > 15, launch_angle < 45) |>
  ggplot(aes(x = launch_speed, y = launch_angle, color = HRprob)) +
  geom_point() +
  scale_color_gradient("HR prob", low = "lightgray", high = "blue")  +
  theme_minimal()
```



That's not really pretty, let's try a grid.

```r
grid <- expand.grid(launch_speed = seq(90, 115, length.out = 50),
                    launch_angle = seq(15,45,length.out = 50))

grid_predictions <- fit |>
  augment(type.predict = "response",
          newdata = grid) |>
  rename(HRprob = .fitted)

grid_predictions  |>  ggplot(aes(x = launch_speed, y = launch_angle)) +
  geom_tile(aes(fill = HRprob)) +
  geom_contour(aes(z = HRprob), breaks = seq(from = 0.1, to = 0.9, by = 0.2)) +
  xlab("Exit velocity (mph)") +
  ylab("Launch angle (degrees)") +
  scale_fill_gradient("HR prob", low = "lightgray", high = "blue")  +
  theme_minimal()
```



Use the 2015 model to predict on 2017.

Obtain all pitches in 2017 in which the ball was put in play.

```r
hit_into_play2017 <- read_csv("statcast_2017_bip.csv") |>
  filter(launch_speed < 150,
         launch_speed > 0,
         launch_angle < 360)

hit_into_play2017 <- fit |>
  augment(type.predict = "response", newdata = hit_into_play2017) |>
```

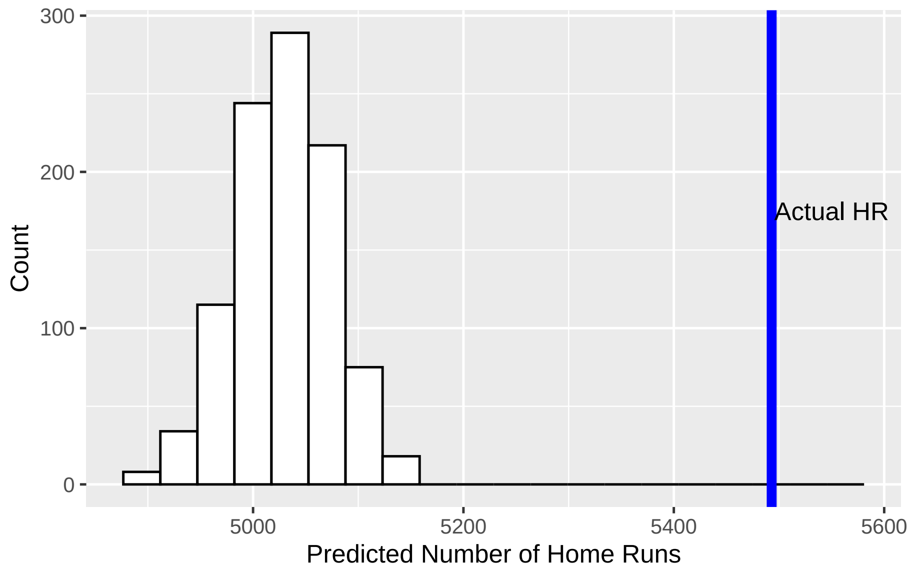
```r
  rename(HRprob = .fitted,HRprobse = .se.fit)

hit_into_play2017 |>
  select(launch_speed, launch_angle, HRprob) |>
  head(20) |>
  knitr::kable()
```

launch_speed	launch_angle	HRprob
95.6	1	0.0000000
107.0	-8	0.0000000
74.3	0	0.0000018
79.6	83	0.0000000
82.7	17	0.0000551
74.8	57	0.0000000
82.9	-21	0.0000000
82.9	-21	0.0000000
100.0	33	0.4157934
60.0	-34	0.0000000
82.9	-21	0.0000000
93.6	55	0.0000024
84.9	-33	0.0000000
46.8	-45	0.0000000
99.6	-4	0.0000000
100.1	-5	0.0000000
78.7	9	0.0000173
85.7	28	0.0013508
93.6	25	0.0249310
76.1	29	0.0001877

Simulate 1,000 seasons.

```r
sim_hr <- function(){
  N <- nrow(hit_into_play2017)
  sum(runif(N) < hit_into_play2017$HRprob, na.rm = TRUE)
}

HR_predict <- tibble(HR = replicate(1000, sim_hr()))

HR_actual <- hit_into_play2017 |>
  count(events) |>
  filter(events == "home_run") |>
  pull(n)
```

```
HR_predict |> ggplot(aes(HR)) +
  geom_histogram(color = "black", fill = "white", bins = 20) +
  geom_vline(xintercept = HR_actual, color = "blue", size = 2) +
  annotate("text", x=5550, y=175, label = "Actual HR") +
  labs(x = "Predicted Number of Home Runs",
       y = "Count")
```



What do the results of this simulation tell us?

```
fit2017 <- hit_into_play2017 |>
  gam(events == "home_run" ~ s(launch_speed, launch_angle),
      family = binomial,
      data = _)


hit_into_play2017 <- fit2017 |>
  augment(type.predict = "response", newdata = hit_into_play2017) |>
  rename(HRprob2017data = .fitted, HRprob2017se = .se.fit)
```

```r
hr_prob_change_fig <-
hit_into_play2017  |>
  mutate(model_diff = HRprob2017data-HRprob) |>
  # filter(!between(model_diff,-.05,.05)) |>
  ggplot(aes(x = launch_speed, y = launch_angle, color=model_diff)) +
  geom_point(alpha=1)+
    # geom_tile(aes(fill = model_diff)) +
  # geom_contour(aes(z = model_diff), breaks = seq(from = 0.1, to = 0.9, by = 0.2)) +
  xlab("Launch speed (mph)") +
  ylab("Launch angle (degrees)") +
  scale_color_gradient2("HR Prob Delta", low = "gray",mid='white', high = "blue",midpoint =
  theme_minimal()

library(patchwork)
hr_prob_2015_fig + hr_prob_change_fig
```